Saliency-Driven Gaze Simulation with Joint Embedding Predictive Architectures

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Abstract

This report reverse-engineers SaccadeJEPA, an unpublished JEPA-style model that uses randomly sampled crop pairs as a self-supervised pretext task, and asks whether its random gaze-target selection can be replaced by a principled saliency predictor. Four saliency sources are evaluated against ground-truth fixation data from 39 observers in the FIND social-video dataset: random baseline, parametric center bias, spectral-residual saliency, and a ResNet-18 model trained on FIND fixation heatmaps. The learned model achieves AUC = 0.891, NSS = 2.549, and CC = 0.713, reducing the gap to perfect gaze prediction by 78 % over SaccadeJEPA's random baseline (AUC = 0.499). ResNetSaliency is designed as a pluggable SaliencySource component; the full gaze-loop integration is addressed in a complementary report.

1. Introduction

Understanding where humans look is a central problem in natural interaction research. Gaze patterns, sequences of fixations and saccades, reveal what information the brain prioritizes and are fundamental to applications in human-computer interaction, robotics, and cognitive science. This report describes the reverse engineering of SaccadeJEPA, a published model that claims to simulate saccadic eye movements, and the development of a principled saliency-based alternative validated against real human eye-tracking data. The question is: can SaccadeJEPA be reused for explicit gaze simulation? This report answers that question directly: not as implemented, because saccade targets are selected uniformly at random. However, by replacing a single line of code with a learned saliency predictor, the resulting system produces gaze patterns that closely match human fixation data.

2. The JEPA Architecture

A Joint Embedding Predictive Architecture (JEPA), proposed by LeCun (2022), learns representations by predicting the embedding of one part of the data from another. Unlike masked autoencoders that reconstruct pixels, JEPA operates entirely in latent embedding space. This makes it more efficient and encourages the model to learn abstract, semantic representations rather than low-level pixel statistics. I-JEPA (Assran et al., 2023) applies this to images using a Vision Transformer (ViT) backbone. The model masks certain patches of an image, encodes the visible context, and predicts the embeddings of the masked regions. The target encoder is maintained as an exponential moving average (EMA) of the context encoder, providing stable training targets without requiring negative samples.

3. SaccadeJEPA: Reverse Engineering

SaccadeJEPA (LumenPallidium, 2023) extends the JEPA framework to simulate saccadic vision. Instead of masking patches, it creates two views of the same image by applying a random affine transformation — a combination of translation and rotation — to produce a shifted view, and a center crop of the original as the reference. The model learns to predict how the embedding changes under the affine transformation. The key mechanism, found in saccade.py line 66, is:

```
rotations    = torch.empty(b, 1).uniform_(-max_rotation_rad, max_rotation_rad)
translations = torch.empty(b, 2).uniform_(-max_translation_frac,
max_translation_frac)
```

This is the entire 'attention mechanism': a uniform random number generator. There is no content-based saliency, no face detection, no semantic understanding. The affine parameters are encoded with a NeRF-like sinusoidal positional embedding and fed to the predictor as a cue, and training uses a combination of Huber loss, cycle consistency, and VICReg regularization — but none of this changes the fundamental finding: the model never decides where to look based on what is in the image. SaccadeJEPA is a representation learning model that uses spatial jitter as a self-supervised augmentation strategy. The 'saccade' is the augmentation, not the output. It cannot simulate gaze dynamics as described in the professor's assignment.

4. Methodology: Saliency Prediction on FIND

4.1 The FIND Dataset

The FIND (Find-Who-to-Look-at) dataset (Liu & Xu, 2016) contains eye-tracking recordings from 39 observers watching 65 multi-face social videos from YouTube and Youku at 1280×720 resolution. Fixations are stored as .mat files. Videos were split into train (53 videos), validation (6 videos), and test (6 videos) using a seeded random permutation (seed=42). Splitting by video rather than by frame prevents temporal data leakage.

4.2 Ground Truth Heatmaps

Raw fixation coordinates from all observers are converted to a spatial probability distribution by placing a 2D Gaussian (σ = 20px) at each fixation location and normalizing the sum to 1. This produces a (224×224) heatmap per frame representing the collective gaze density.

4.3 Saliency Sources Compared

• Random saliency: A uniform random map normalized to sum to 1. This directly replicates SaccadeJEPA's behavior.
• Center bias (parametric): A hand-crafted 2D Gaussian centered on the image.
• Spectral residual (classical): Implemented using OpenCV's StaticSaliencySpectralResidual (Hou & Zhang, CVPR 2007). A bottom-up, non-learned signal-processing approach that identifies salient regions from local spectral deviation — no knowledge of faces or semantics.
• ResNetSaliency (learned): A pretrained ResNet-18 backbone with a lightweight convolutional head trained on FIND fixation heatmaps.

4.4 ResNetSaliency Architecture

The model uses a ResNet-18 backbone pretrained on ImageNet, keeping layers up to layer3 (14×14 feature maps at 224×224 input). The backbone is frozen during training to preserve pretrained BatchNorm statistics. The head consists of two 1×1 convolutions (256→64→1 channels) with GELU activation; logits are upsampled to 224×224 via bilinear interpolation and normalised with spatial softmax. Total parameters: 2,799,297. Trainable parameters: 16,513 (head only). Training: 20 epochs, batch size 16, AdamW (lr = 1×10^{-3}, weight decay = 1×10^{-4}), cosine annealing schedule, KL-divergence loss computed at 14×14 resolution (196 bins — computing KL at full 224×224 = 50,176 bins is intractable with batch size 16). No image augmentation was applied beyond resizing to 224×224.

5. Experiments and Results

5.1 Evaluation Metrics

Three standard metrics from Bylinskii et al. (2019) are used. AUC (Judd version) treats each fixation as a positive sample and sweeps a threshold over the saliency map; 0.5 is chance, 1.0 is perfect. NSS (Normalized Scanpath Saliency) normalizes the map to zero mean and unit variance, then reads the value at fixation locations; 0 is chance, and a value of 2.5 means fixated pixels score 2.5 standard deviations above the map average. CC (Pearson Correlation Coefficient) measures spatial agreement between the predicted map and the ground-truth fixation density; 0 is no correlation, 1 is perfect.

Saliency Source	**AUC ↑**	**NSS ↑**	**CC ↑**
Random (SaccadeJEPA baseline)	0.499	-0.004	0.000
Center bias (parametric)	0.774	1.056	0.460
Spectral residual (classical)	0.687	0.391	0.132
ResNetSaliency (ours)	0.891	2.549	0.713

5.3 Discussion and Limitations

The random baseline achieves AUC = 0.499, confirming chance-level performance and validating the evaluation pipeline. ResNetSaliency achieves AUC = 0.891, NSS = 2.549, and CC = 0.713, outperforming all baselines on every metric. Much of this power comes from ImageNet pretraining, which implicitly encodes face and person detection — the dominant visual content in FIND social videos. Results are reported as point estimates averaged across 310 test frames; no standard deviations, confidence intervals, or per-video breakdowns are provided, which limits conclusions about robustness to individual video variation. Limitations: the training split is small (53 videos), temporal dynamics across frames are not modelled, and the model is unlikely to generalise beyond face-heavy social video without a more diverse training dataset.

6. Can SaccadeJEPA Be Reused for Gaze Simulation?

Yes — with one change. Replacing the random uniform line with saliency-guided sampling:

```
saliency     = ResNetSaliency(frame)
fixation     = sample_from(saliency)
translations = fixation_to_ndc(fixation)
```

The improvement is substantial. The gap to perfect AUC prediction is reduced by 78%, computed as $(0.891 - 0.499) / (1 - 0.499) = 0.392 / 0.501 \approx 0.782$. Because ResNetSaliency follows the SaliencySource interface, it slots directly into this position as a content-aware prior. The full gaze-loop integration — sequential fixations, inhibition-of-return, and the cropper substitution — is outside the scope of this report (see Emad Bahreinipour's complementary report for the complete GazeCropper implementation).

7. Conclusion

SaccadeJEPA is a representation learning model, not a gaze simulator. Its saccade targets are uniformly random, producing fixation patterns indistinguishable from chance (AUC = 0.499, NSS = -0.004). By replacing that single random line with a ResNet-18 saliency predictor trained on FIND eye-tracking data, we obtain a content-driven gaze model that achieves AUC = 0.891, NSS = 2.549, and CC = 0.713 on held-out social videos — a 78% reduction in the gap to perfect gaze prediction. The learned model's strength reflects the face-heavy structure of FIND; a more diverse training set and full backbone fine-tuning would yield broader generalisation.

References

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